SE Lab 3: Architecture Justification

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Problem 64: Disaster Relief Supply & Volunteer Coordinator

Architecture Selection: We chose the **Microservices Architecture** for the Disaster Relief System.

Architectural Choice Justification:

1. **Independent Scaling:** During a live disaster, certain system components experience massive, sudden spikes in traffic. For example, the Live Disaster Mapping Dashboard and GPS Tracking Service will handle thousands of concurrent requests as volunteers move and managers coordinate routes. A microservices architecture allows us to independently scale these high-load routing services without unnecessarily scaling the Shelter Supply Inventory, saving resources and preventing total system failure under load.
2. **Fault Isolation:** In an emergency response environment, system resilience is critical. If the Emergency Alert System experiences a failure or network timeout due to third-party SMS gateway issues, a microservices approach ensures this failure is isolated. The Skill based Volunteer Dispatch and the Live Disaster Mapping Dashboard will continue to function normally, allowing the Disaster Manager to keep assigning volunteers manually while the alert service recovers.

Security Advantage:

Microservices provide a distinct security advantage by allowing granular access control and isolated data perimeters. The Live Disaster Mapping Dashboard does not have direct access to the underlying Shelter Supply Inventory database. Instead, it must communicate through a secure API gateway, minimizing the attack surface and preventing unauthorized database queries.

Performance Benefit: This architecture inherently reduces network latency by allowing independent deployment and optimized technology stacks for specific tasks. For example, the GPS Tracking Service would be built using a lightweight, high-throughput protocol for real-time location streaming, while the Shelter Supply Inventory could utilize a traditional, ACID-compliant relational database for accurate stock counting.